BowelProbe: Real-Time Bowel Viability Assessment Using Electrical Impedance Spectroscopy

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1. Problem Description

Acute bowel ischemia remains one of the most lethal surgical emergencies, with **mortality rates reaching 60–80%** for acute mesenteric ischemia [1], [2]. Surgeons must accurately distinguish viable from non-viable tissue to avoid two catastrophic outcomes: under-resection, leading to sepsis, or over-resection, leading to Short Bowel Syndrome (SBS).

Intraoperative decision-making is fraught with uncertainty; research indicates that up to **25% of patients** with suspected ischemia undergo "second-look" operations to reassess borderline tissue [3]. These repeat surgeries significantly increase patient risk and add an average of **$25,000 to hospital costs per encounter** [4]. Conversely, aggressive resection results in SBS, which carries an annual treatment cost exceeding **$150,000 per patient** due to lifelong parenteral nutrition [5]. By providing objective viability data, *BowelProbe* directly addresses this multi-billion dollar clinical burden.

2. Solution Concept

Impedance-Based Assessment

We propose *BowelProbe*, a handheld, sterilizable device using Electrical Impedance Spectroscopy (EIS) to quantify tissue health. Ischemia disrupts cellular membrane integrity and ion gradients, causing measurable decreases in electrical impedance. Unlike dye-based imaging, EIS requires no consumables, allows for continuous measurement, and directly reflects structural cellular status rather than just perfusion.

Supporting Evidence

Hou et al. demonstrated that EIS can detect ischemic injury in human small intestines, showing a progressive decline in impedance over time (Figure 1) [6]. These findings are supported by prior work demonstrating that multi-frequency bioimpedance measurements can reliably characterize tissue structural changes across ischemic states [7]. Porcine studies further confirm that these impedance-based features correlate with histopathologic outcomes and provide a reliable metric for identifying tissue that has surpassed the point of reversible injury [8], [9].

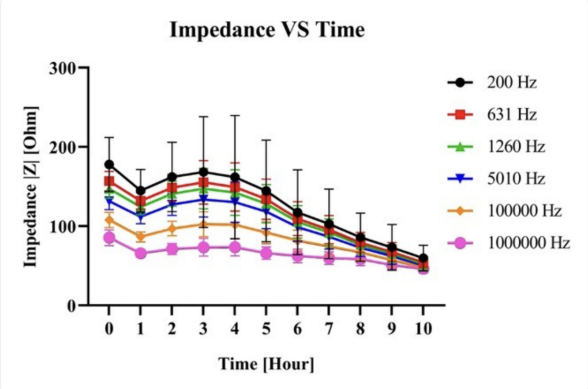


Figure 1: Mean electrical impedance of human small intestine over time. Impedance decreases as cellular membranes degrade. Adapted from Hou et al. [6].

3. Reduction to Practice

Proof-of-Concept Measurement Architecture

The proof-of-concept utilizes a voltage-divider architecture with a 100 Ω precision reference resistor. To ensure signal integrity and prevent voltage drops caused by downstream loading, the voltage across the probe is passed through a high-impedance buffer circuit.

The buffered AC signal is then processed via a "super diode" (precision rectifier). This stage removes the negative half-cycle of the waveform while eliminating the standard 0.7 V diode forward-voltage drop, ensuring high accuracy for low-amplitude biological signals. Finally, an RC low-pass filter smooths the rectified signal into a stable DC voltage, which is digitized by the microcontroller for real-time viability computation.

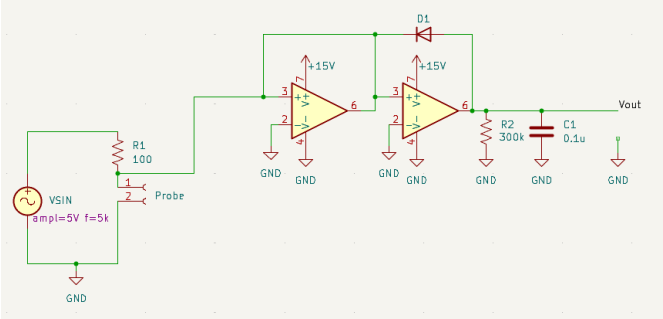


Figure 2: Electronic schematic: Signal chain featuring voltage divider (100 Ω), buffer, precision rectification (super diode), and RC filtering stages.

System Scalability

The production model will integrate all electronics into a compact, handheld module. The architecture supports multi-frequency measurements and noise rejection, allowing it to integrate seamlessly with existing surgical displays. A high level overview of the device can be seen in Figure 3.

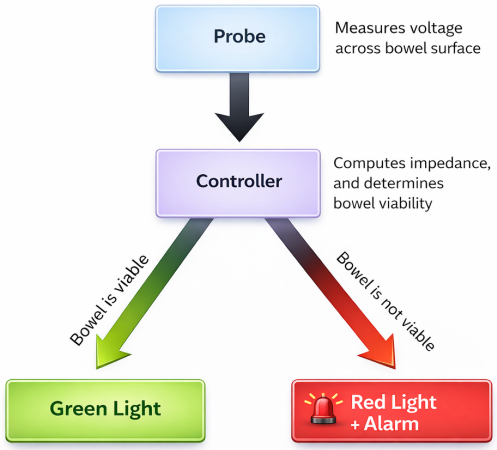


Figure 3: Functional block diagram of the BowelProbe system. The workflow includes real-time signal acquisition, impedance-based viability computation, and dual-mode output for intraoperative decision support.

4. Pathway to Implementation

Regulatory and Intellectual Property

BowelProbe is expected to follow the **FDA 510(k) Class II** regulatory pathway, utilizing existing bioimpedance monitors as predicates (Product Code DSB). A DeNovo submission may also be considered if the FDA requires a PMA submission due to the novel use of the technology. The BowelProbe will undergo rigorous testing to meet **IEC 60601-1** (Medical Electrical Equipment Safety) and **ISO 10993** (Biocompatibility) standards. A preliminary patent search indicates high novelty in the integration of **optimized electrode geometries** and **real-time diagnostic algorithms** for handheld, multi-frequency EIS specifically designed for intraoperative bowel viability assessment.

Validation and Clinical Translation

Technical development will focus on optimizing electrode geometry to ensure uniform contact pressure. Validation will move from benchtop phantom testing to ex vivo animal models, eventually utilizing in vivo ischemia–reperfusion models to compare BowelProbe performance against gold-standard histopathology.

5. Competitive Advantage & Impact

As shown in Table 1, BowelProbe offers a quantitative advantage over current standards. Unlike ICG-FA, which requires expensive consumables and only measures blood flow, BowelProbe assesses the **cellular structural integrity** itself.

Feature	Visual	ICG-FA	BowelProbe
Metric	Subjective	Perfusion	**Viability**
Cost/Use	$0	~$150+	**$0**
Real-Time?	Yes	No (Dye delay)	**Continuous**
Quantitative?	No	Qualitative	**Yes (EIS)**

Table 1: Comparison of intraoperative assessment methods.

Clinical and Economic Impact

BowelProbe directly identifies tissue that is irreversibly injured even when appearing "perfused." This precision reduces the incidence of **anastomotic leak (a $30,000 complication)** and minimizes planned second-look operations. For healthcare systems, this translates to faster ICU throughput and significant reductions in the **$150,000/year** cost burden associated with Short Bowel Syndrome. BowelProbe integrates seamlessly into existing operative workflows, requiring less than five seconds per measurement, no contrast agents, and minimal training. Unlike fluorescence-based systems, it does not require room darkening, additional imaging hardware, or delays related to dye administration.

References

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